

# Stakeholder Engagement Plan

AgriAdviser-NG

## 1. Selected stakeholder groups

Drawing on the stakeholder map and the underrepresentation risks already flagged in the Data Card, this plan focuses on:

- Local agricultural extension officers (NAERLS field agents, State ADP extension staff)
- Women smallholder farmer cooperative groups

## 2. What matters most to them

Extension officers care most about the tool complementing, not undermining, their authority — they want its advice to align with official extension guidance, defer clearly when uncertain, and reduce their workload rather than create conflicting information farmers must reconcile themselves. Women's farmer cooperatives prioritize genuine inclusion despite lower smartphone ownership and literacy rates, privacy when using a shared or borrowed phone, and representation of the specific crop and post-harvest tasks they typically manage, which extension literature often overlooks. Both groups want a system that is trustworthy, transparent about its limits, and shaped by people who actually understand local farming realities.

## 3. Potential harms and benefits

**Extension officers** — Potential harms: farmers bypassing them entirely, the tool contradicting their guidance and eroding trust in either source, or their domain expertise being extracted into the dataset without acknowledgment or ongoing involvement. Potential benefits: reduced repetitive-question burden, a tool that routes farmers to them precisely when human judgment is needed, and a channel to correct misinformation before it spreads.

**Women's farmer cooperatives** — Potential harms: exclusion from the dataset if their specific crops, tasks, or phrasing patterns are underrepresented in the source material; privacy risks if questions sent from a shared phone are traceable to an individual; reinforcing a system that assumes male-typical farming priorities as default. Potential benefits: direct access to guidance without needing to travel to or wait for a male relative or officer, and a dataset finally shaped around their actual questions rather than assumptions about them.

## 4–5. Engagement methods

### Extension officers → Participatory Design

Participatory Design is the right fit for extension officers because their relationship to this project is fundamentally different from an end-user's — they are domain experts whose professional judgment the model is explicitly built to defer to, not simply a population the system serves. A method that treats them as passive subjects to be surveyed would misrepresent that relationship and risk exactly the harm they're most worried about: being extracted from for their knowledge without being genuinely involved in how that knowledge gets used.

Concretely, this would take the form of structured co-design sessions where a small group of NAERLS and State ADP officers directly review real model outputs against real farmer questions, flagging cases where the generated advice is subtly wrong, outdated relative to current extension guidance, or technically correct but phrased in a way that could mislead a farmer without agronomic background. This is a fundamentally different — and more useful — activity

than one-off feedback collection, because it lets officers see the system's actual failure patterns across many examples, not just react to a single output in isolation.

Participatory Design also directly addresses the “authority erosion” harm identified above. If officers are only consulted after the system is built, their reasonable fear — that farmers will trust a chatbot over a human whose training and local knowledge the chatbot can't fully replicate — has no outlet except distrust or resistance to the tool's rollout. Involving them in shaping the system's abstention behavior specifically (when should it say “I'm not confident, contact your extension officer” rather than answer) gives them direct influence over the exact mechanism meant to protect their role, rather than being told after the fact “don't worry, it defers to you.”

This method also has a practical benefit for the project's use of official sources. NAERLS and ADP officers are the people best positioned to tell us which extension bulletins are current, which have been superseded, and which recommendations carry important regional caveats that a static PDF extraction might miss or flatten. A participatory design session isn't just a trust-building exercise — it's a genuine quality-control layer that no amount of automated dataset validation can substitute for, since it requires domain judgment about whether a piece of advice is still correct in practice, not just internally consistent with its source document.

Finally, this method fits realistically within how extension officers already work. Co-design workshops resemble the kind of professional development or peer-review sessions officers already participate in through their institutional structures, meaning the engagement format itself doesn't impose an unfamiliar burden — it slots into an existing professional practice rather than asking for an entirely new kind of participation. Given the project's explicit commitment to augmenting rather than replacing extension work, this alignment between the engagement method and officers' existing professional context is itself part of what makes the method appropriate, not just convenient.

## Women's farmer cooperatives → Distributed Dialogue

Distributed Dialogue is the better fit for women's farmer cooperatives precisely because the barriers that risk excluding them from this project — limited mobility, time constraints from unpaid domestic and farm labor, lower rates of independent travel to a central meeting point — are the same barriers a single centralized workshop would reproduce. A method requiring participants to travel to one location at one time systematically filters out exactly the people most at risk of being underrepresented in the dataset already, defeating the purpose of the engagement before it starts.

Distributed Dialogue is designed for exactly this situation: gathering structured input across multiple smaller, geographically dispersed conversations, often facilitated locally rather than centrally, and then synthesizing themes across them rather than requiring convergence in a single room. In practice, this would mean working through existing cooperative structures — meeting women's groups where they already gather, at times that fit around their existing labor and childcare responsibilities, facilitated where possible by someone from their own community rather than an outside researcher, which also helps address the comfort and trust concerns tied to discussing farming challenges (and by extension, household economic vulnerability) with strangers.

This method directly targets the representation harms identified earlier. If the dataset's current agronomic source material skews toward crops and tasks documented by formal (often male-dominated) extension literature, Distributed Dialogue conversations are where questions about the specific crops, livestock, and post-harvest tasks women's cooperatives actually handle can surface — information that a synthetic-question-generation process, however careful, cannot invent on its own, because it can only paraphrase what's already in the source material. This is the direct link between engagement and the “no source, no answer” principle underlying the dataset's design: expanding the range of authentic questions collected through dialogue is what eventually lets more of the dataset shift from synthetic to genuinely sourced.

The privacy dimension also shapes why this method, done well, matters more than it might first appear. A distributed, locally facilitated dialogue allows a level of contextual sensitivity — understanding, for instance, that a phone used to access the system might be shared within a household — that a generic survey or centralized session is less likely to surface, because it depends on trust built through a familiar facilitator and setting rather than a one-time interaction with an unfamiliar research team.

Finally, this method scales naturally with the project's broader commitment to auditing geographic and demographic coverage rather than assuming it. Distributed Dialogue sessions run across multiple cooperative groups in different

regions double as a practical mechanism for checking whether the dataset's agro-ecological zone coverage — already flagged as a known gap — actually reflects where these specific communities farm, rather than relying on whichever zones happened to be best documented in existing extension literature.

## 6. How this feedback would shape the dataset and model

Extension officer co-design sessions would directly refine the abstention threshold — teaching the model when a question genuinely needs a human referral rather than a generated answer — and flag outdated or regionally inaccurate source material before it enters training. Distributed Dialogue with women's cooperatives would surface authentic question phrasing to replace synthetic Pidgin approximations, and reveal specific crops, tasks, or agro-ecological zones currently missing from source coverage. Together, both engagements would shift the dataset's origin field balance further toward genuinely sourced content, and directly inform which topic and crop gaps the next sourcing round should prioritize.